ANUSHKA JHA *she/her*

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Summary

AI/ML Engineer pursuing a Master's in Computer Science at Arizona State University with hands-on experience developing LLM applications, computer vision models, and end-to-end machine learning pipelines. Skilled in Python, PyTorch, Transformers, and Retrieval-Augmented Generation (RAG), with a passion for building practical AI solutions that solve real-world problems.

Education

Arizona State University *Tempe, Arizona*
Masters in Computer Science *Aug 2025 – May 2027*
SRM Institute of Science and Technology *India*
Bachelor of Technology in Computer Science and Engineering (AI and ML specialization) *Aug 2021– May 2025*
Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Data Mining, Data Structures & Algorithms, Database Management Systems, Probability & Statistics, Linear Algebra

Technical Skills

Programming : Python, C++, C, Java
AI / Machine learning : PyTorch Deep Learning Machine Learning Transformers LLMs Natural Language Processing Computer Vision Prompt Engineering Structured Outputs Retrieval-Augmented Generation (RAG)
DataBase : SQL, Relational Database, Data processing, Batch processing
Tools : Git, Linux, VS code, CodeEx, Claude workspace, Cursor

Work Experience

Machine Learning Intern ***Nov 2024 – May 2025***
Fxis.AI

- Built a Python pipeline that extracted structured candidate information from resumes using the ChatGPT API, reducing manual resume screening effort and achieving approximately 85% extraction accuracy through manual evaluation.
- Designed prompts and validation logic to extract information consistently from resumes with different layouts while preventing missing information from being filled with incorrect or inferred data.
- Developed a recommendation system that ranked relevant job content using SQL queries and ranking logic, increasing recommendation engagement by approximately 20–25%.
- Automated candidate communication by integrating the Twilio WhatsApp API with the recruitment workflow, maintaining reliable message delivery for hiring campaigns.

Machine Learning Intern ***May 2024 – June 2024***
Prodigy InfoTech

- Developed a convolutional neural network (CNN) for hand gesture recognition using PyTorch and OpenCV, contributed to a vision system that classified hand gestures with over 90% test accuracy.
- Prepared and preprocessed image datasets by resizing, normalizing, and augmenting training images, improving model robustness across different lighting conditions and hand positions.
- Worked closely with the development team to integrate the trained model into the overall gesture recognition application and validate its performance on real image samples.

Projects

Gait Recognition and Classification using Deep Learning | *Research Paper (Publication in review)*

- Developed an end-to-end deep learning pipeline for gait recognition and classification using the CASIA-B dataset (~500K images across 11 camera views), achieving over 85% test accuracy and 98% validation accuracy.
- Evaluated transfer learning models including VGG16, InceptionV3, ResNet101, and EfficientNetB2 with LSTM, BiLSTM, CNN-LSTM, CNN-BiLSTM, and 3D CNN architectures to identify the best-performing model for gait analysis.
- Improved model performance through image preprocessing, data augmentation, feature extraction, PCA-based dimensionality reduction, and hyperparameter tuning, and evaluated results using precision, recall, F1-score, and confusion matrices.
- Built separate pipelines for gait classification (walking condition) and gait recognition (person identification), demonstrating the effectiveness of deep learning for multi-view biometric recognition.

Various Machine Learning Model for Water Quality Classification| *Published Research Paper*

- Developed an end-to-end machine learning pipeline to classify water potability using physicochemical features, supporting automated water quality assessment for environmental monitoring.
- Prepared and analyzed the dataset through exploratory data analysis (EDA), mean imputation for missing values, outlier handling, feature scaling, and train-test splitting to improve data quality before model training.
- Evaluated Support Vector Machine (SVM), Decision Tree, and XGBoost models using grid search and performance metrics including accuracy, precision, recall, F1-score, and confusion matrices, with XGBoost achieving the best classification accuracy of 93.07%.
- Analyzed feature relationships and model behavior to identify key indicators of water quality, improving model interpretability and supporting data-driven model selection.